SITEWISE AI · PROJECT REVIEW

SiteWise AI

AI-Powered Land Suitability Analysis

End-to-End Website Workflow Using U-Net



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SYSTEM ARCHITECTURE

From uploaded image to land-use insight

SiteWise AI turns an uploaded satellite or aerial image into preliminary land-use decision support.



The workflow keeps validation, model inference, interpretation, and reporting connected.

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01 · INGESTION

The workflow starts with a validated upload

01 Open SiteWise AI

Select image analysis.

02 Upload imagery

Choose a satellite or aerial image.

03 Validate the file

Check format, size, and dimensions.

04 Send the approved file

Pass it to the Flask backend through an API.

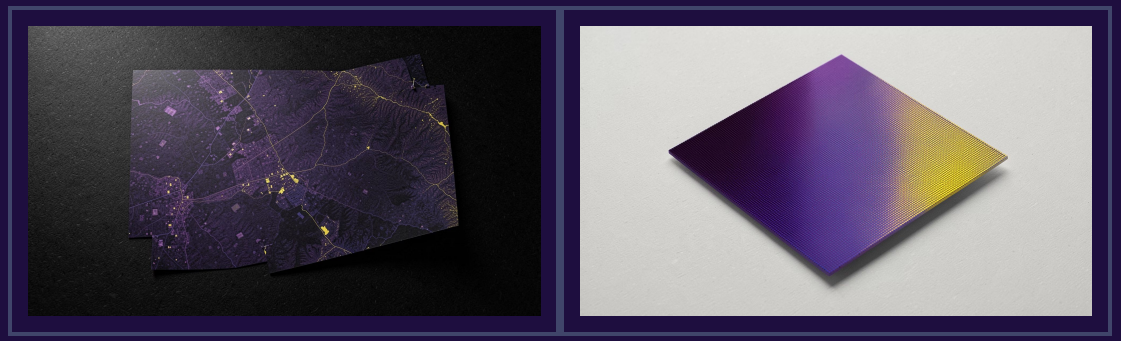


The API receives only an approved image payload.

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02 · PREPROCESSING

Preprocessing makes the image model-ready



Original scene → consistent model-ready representation

01 Resize

Match the model's required input dimensions.

02 Convert

Bring channels into RGB.

03 Normalize

Scale pixel values for stable inference.

04 Return tensor

Use one consistent contract downstream and during training.

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03 · QUALITY GATE

Reliability is checked before results are trusted

The Reliability CNN adds a quality signal alongside segmentation.

01

Uploaded image

Quality enters the workflow with the file.

PASS

Proceed with the analysis.

02

Reliability CNN

Evaluates degradation and poor-quality inputs.

CAUTION

Interpret downstream output carefully.

03

Quality signal

Pass, caution, or review state travels with the result.

REVIEW

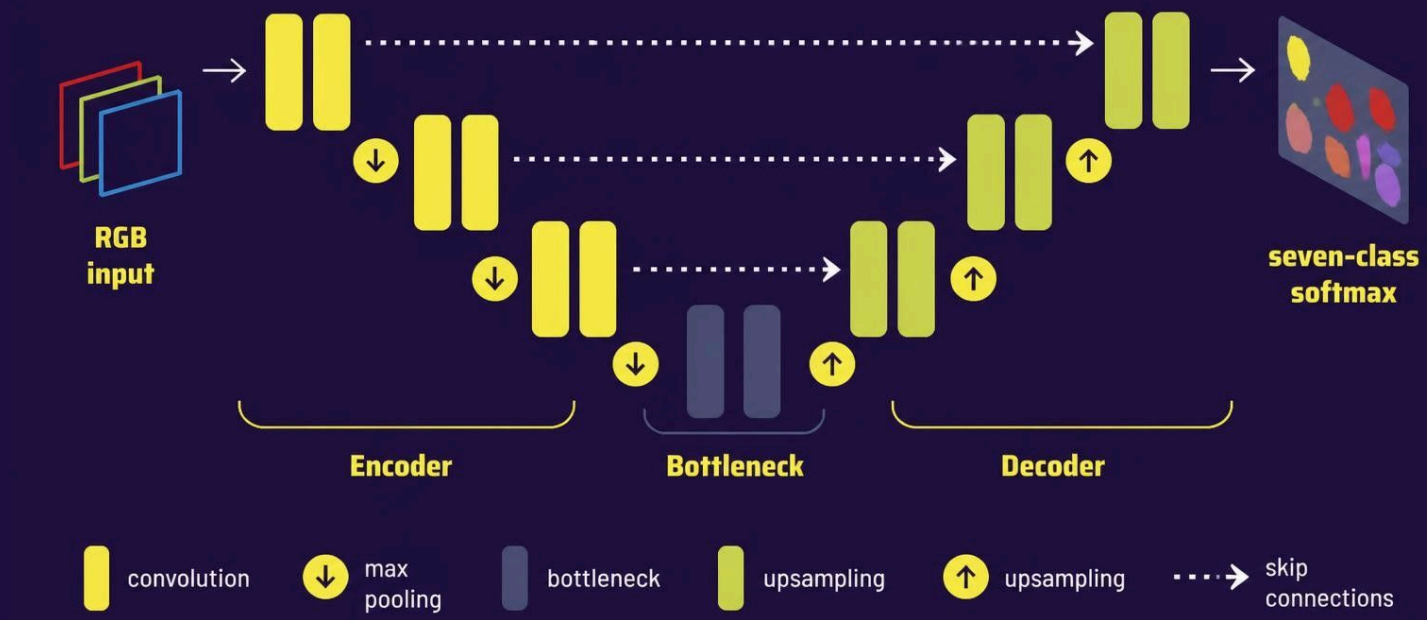
Quality is too weak to trust without review.

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04 · SEGMENTATION MODEL

U-Net links context with pixel precision

Encoder features move toward deeper context; skip connections return the spatial detail needed for a precise mask.

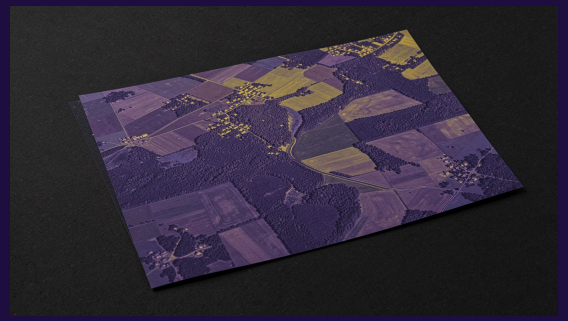


Final output: a seven-class softmax prediction for each pixel.

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05 · PIXEL CLASSIFICATION

Every pixel receives a land-cover class



Preprocessed RGB scene



Coloured segmentation mask

Class assignment

U-Net predicts a probability distribution for every pixel. The most likely class becomes the pixel's land-cover label.

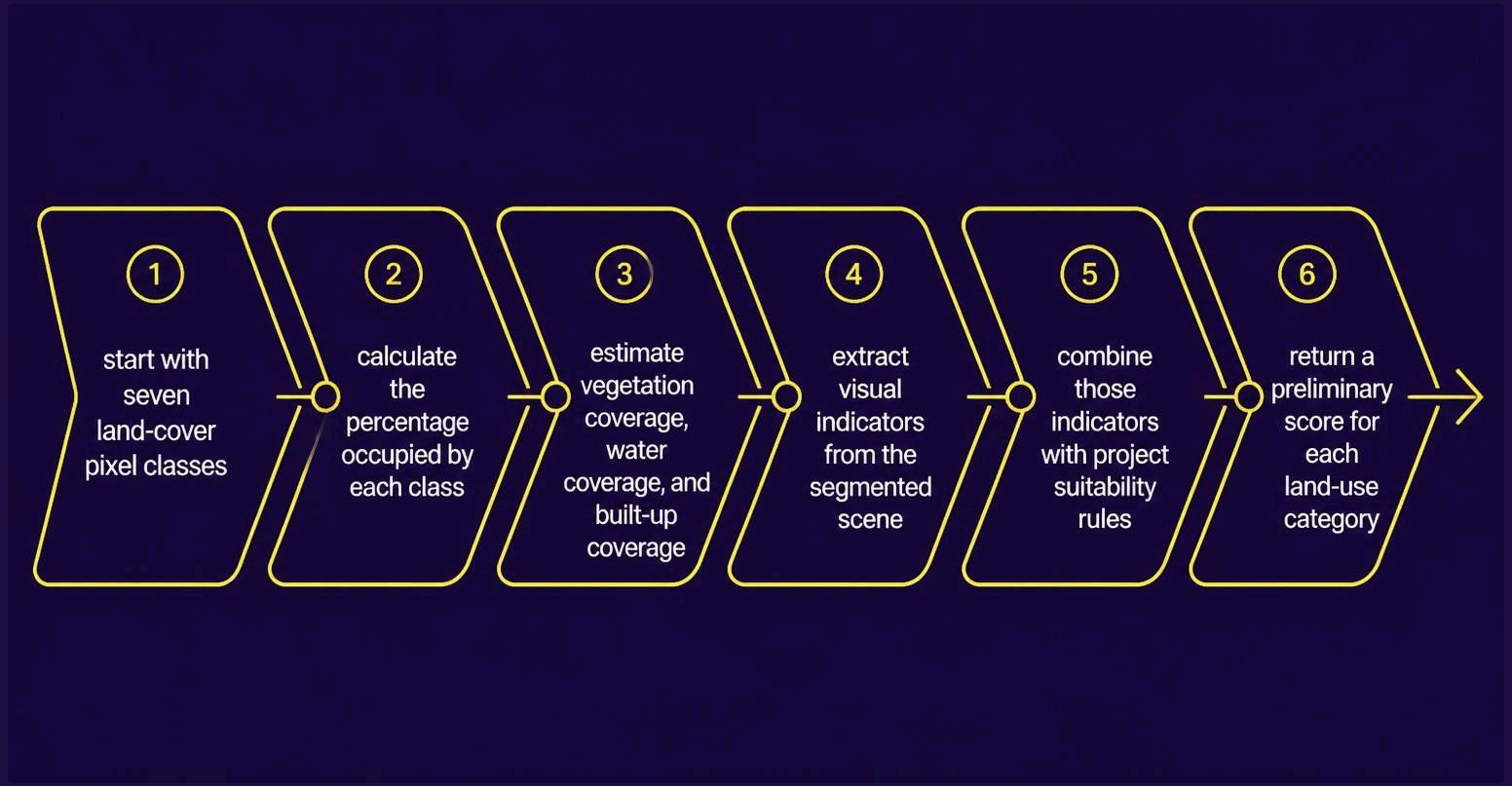
Urban	Agriculture	Rangeland
Forest	Water	Barren
Unknown		

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06 · INTERPRETATION

Features become preliminary suitability scores

The segmented scene becomes structured evidence for the suitability engine.



No project-specific thresholds or final scores are assumed here.

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07 · ORCHESTRATION AND DELIVERY

Flask connects processing to the results dashboard

01 `browser`

Upload the validated image.

02 `Flask API`

Coordinate the processing request.

03 `preprocess`

Prepare the model-ready tensor.

04 `Reliability CNN · U-Net`

Assess quality and generate the segmentation.

05 `features · scoring`

Aggregate indicators and suitability outputs.

RESULTS DASHBOARD

Returned outputs

Mask + class legend

Metrics + reliability status

Suitability score summary

Structured results return to the website for review.

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08 · REPORTING

The final report preserves the full analysis trail

Upload

Record the source image.

Process

Summarize preprocessing and inference.

Interpret

Include the mask and land-cover breakdown.

Review

Record reliability findings and interpretation boundaries.

Report

Present preliminary suitability scores.

Report contents: uploaded image summary · processing outcome · segmentation output · land-cover breakdown · reliability findings · suitability scores

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